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MILESTONE 18

JOINT SPECIFICATIONS OF SPARE PARTS DATABASE READY

Work Package #	WP3: Ramping-up Towards Excellent Steady-State Operations
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Abstract:

Reaching the current milestone has demonstrated the significant progress achieved in the first year of the project. Several iterations of joint work and collaboration resulted in that the first specification for the spare parts database is ready. A hybrid workshop held on 12-13 October 2021 has supported significantly to achieve the milestone.

This document describes the first version of the specifications, especially focusing on the following topics: risk management aspects, conceptual and technical approaches, as well as requirement specifications. It is important to highlight that it is only the first step in specifying the database as a more detailed specification including a development roadmap will be the next step and is due by April 2022.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update and Delete (basic operation of persistent storage)
AGILE	An iterative approach to project management and software development
DTFO	Delivery time from order
EDSP	ELI Database of Spare Parts
ELI	Extreme Light Infrastructure
ELI ERIC	ELI European Research Infrastructure Consortium
ELI-ALPS	ELI Attosecond Light Pulse Source Facility
ELI-NP	ELI Nuclear Physics Facility
ELI-PANDA	ELI oPerations And maiNtenance DAtabase
ERP	Enterprise Resource Planning
GA	Grant Agreement
GitHub	Git respository and continuous integration provider
GUI	Graphical User Interface
GUID	Globally Unique IDentifier
LDAP	Lightweight Directory Access Protocol
MLDT	Mean Logistics Delay Time
MTTR	Mean Time to Repair
On-prem	On-premises software is installed and runs on computers on the premises of the person or organization using the software, rather than at a remote facility such as a server farm or cloud.
OPEX	O perational e xpenditure
REST	Representational State Transfer (architectural style)
RUL	Remaining useful time
VM	Virtual machine
VS Code	Visual Studio Code Editor

1 Introduction and goals

The premise that good maintenance practices are fundamental to success is beyond question. In accordance with IMPULSE Project requirements, ELI facilities had an obligation to create a joint spare parts database. The essential intention behind this requirement had several purposes, such as: to build up a relevant database in order to minimize possible downtime for user experiments, to determine which spare parts must be stocked in advance, to make cost-effective maintenance decisions, just to name a few.

All ELI facilities have developed and commissioned extremely complex state-of-the-art laser and laser-driven technologies through a combination of external contracts with suppliers and internal in-house development. These technologies represent a unique, purpose-built system with many custom critical components with long lead times. The reliable operation and maintainability of such complex systems is highly dependent not only the timely availability of spare parts but also on the availability of well-trained experts capable of servicing these systems. The facilities, therefore, need to have well thought out plans and procedures on how to maintain such complex systems and analyse for each subsystem whether the maintenance is most efficiently done with in-house available spare parts and in-house available expertise and tools or whether it is more effective to outsource some of the maintenance as a service.

The reliability of the systems and “operation ease” has been considered during the development stage and is reflected in the choice of reliable components as well as inbuilt machine safety functions that protect the systems from unwanted damage and its propagation to the linked subsystems. There are, of course, limits to what machine safety and diagnostic systems as well as automation can do to protect the whole system from all possible damage scenarios. Improving equipment reliability can be one of the aspects that can significantly contribute to achieving effective maintenance. However, it takes a holistic approach that includes equipment performance as a major enhancement to productivity. In an attempt to have ELI facilities available as much as possible for users, our experience has taught us that the rule of thumb is the elimination of failures. Uptime of equipment is what counts and what we are aiming for.

The ELI database of spare parts (EDSP) to be developed within IMPULSE task 3.3 should first of all record the in-house available spare parts, but it should also be a tool that potentially helps to analyse the complex structure of the laser and experimental system and interdependences between them and then identify critical equipment. It also needs to analyse the criticality of each in-system part in order to determine which spare parts must

be stocked or which service contracts need to be in place in order to minimize downtime and impact on operation of user experiments.

The EDSP should help to minimize the downtime in case of any “in-system part” failure. It should help to identify which “in-system parts” will have the most efficiency preventing the system downtime in case of failure. The EDSP should therefore enable staff to carry out the criticality analysis of the “in-system parts” (categorizing them into several criticality classes) based on multiple criteria and guiding systematic purchasing of most critical spare parts so that they are available in stock at ELI or in stock at a supplier (guaranteed by a supply contract). This process is described below in 2.5.

The EDSP shall record in detail the parameters of the spare parts in order to enable easy search for suitable replacement parts by its parameters as described in 2.6.

The EDSP should also help to maximize reliability and cost efficiency through tools providing:

- means of sharing critical spares between multiple systems,
- support of parts standardization,
- spare parts usage statistics
- interfaces to procurement systems etc.

According to the task description, the database should be specified jointly and then implemented at each Facility separately, while ensuring data extraction features to allow for joint review and analysis. However, after several iterations and versions of the conceptual architecture, all the partners agreed to cooperatively develop one common database for all three ELI sites also with common front-end GUI and common access management through user permissions and pre-defined roles of users (see Figure 1). This common implementation approach would be more advantageous in terms of cost-effectiveness, shared codebase, shared cloud-based infrastructure, and so forth.

Given the historical availability of various data sets at ELI-ALPS, ELI-NP and ELI-Beamlines related to system structure, procurement, risk analysis and spare parts, the development of EDSP must include import tools and/or generic interfaces to import these historical data sets. This local adaptation of existing data sources for the new common database will be handled through locally developed migration software that will connect to EDSP gateway using the beforementioned import tools & interface (OPEN API). Similarly, centrally and/or locally implemented ERP systems can interface with the EDSP through the same gateway.

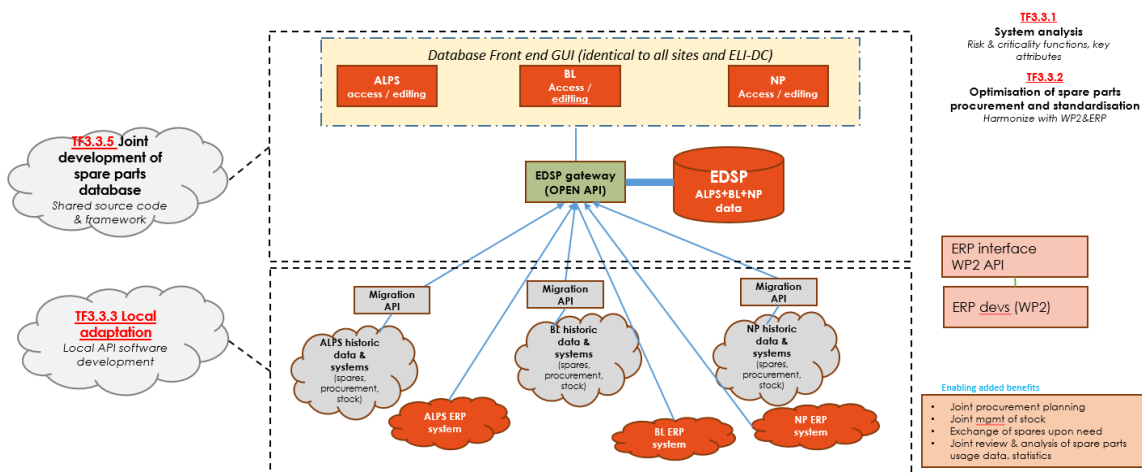


Figure 1: Conceptual diagram describing the shared database for all three sites with separate access to local data and methods for migrating data from existing local data sources.

It should be noted that this document represents initial thoughts on the approach to spare parts management and optimization and is likely to significantly develop over time, i.e., the detailed specifications of the spare parts database and development roadmap will be provided as a complete design reference in the form of a deliverable, D3.8, due by April 2022. The immediate need is for the database to support spare parts management, consolidation and standardization. However, in the long term, it is expected that the database will develop and assume more functionality to support operation and maintenance in general including for example equipment service logs, planning of preventative maintenance etc. The overall goal of spare parts optimization is to maximize our operational availability for users while minimizing life cycle costs.

Although the EDSP is primarily developed to optimize stocking of spare parts that are needed to maintain the high availability of the ELI technology for user experiments, its functionality can be further extended to also support system maintenance.

Maintenance practices such as regular planned maintenance, condition monitoring (CM), preventative maintenance as well as personnel education and training, systems for planning and controlling work, expert systems etc. are considered essential to achieving world-class performance. Not all of these topics can be covered by this database but, if integrated with other modules of the ERP system and project management software it can have a significant impact on the availability and reliability of the laser and secondary beams offered to the ELI users.

Briefly summarizing the above thoughts, in long term, the ELI Database of Spare Parts (EDSP) could evolve to integrate into a future ELI operations and maintenance management system.

1.1 Risk management

This subsection briefly discusses the potential risks of Task 3.3 specification and implementation.

The following list summarizes the key potential risks:

- potential delay, late delivery of central functionalities (or even not delivering)
- potential delay and late delivery of local functionalities, i.e., data migration (or even not delivering)
- improper and/or incomplete integration with other central/local systems (especially ERP systems)

The common and major root causes for the above risks are the following:

- Scope creep, users need to stay focused on the current scope of the project, focusing on the development of a spare parts database. As presented by the Kano model, the teams can be easily defocused trying to develop the next-level, better version of a framework even before they have the first functional product in place.
- Lack of the necessary required IT and/or other expertise, following the scope creep effect described above, in case of a very ambitious goal is set, we need to also take into account that the development is to be done by a few software developers, which is not in accordance with all development ideas already identified and discussed.

The common and major prevention and/or mitigation barriers for the above-described risks and major root causes are the following:

- Follow an iterative, incremental software development, an AGILE approach, i.e., each iteration shall deliver working and deployed solutions for the respective relatively small, gradual and incremental step in terms of functionality
- Only the basic and essential requirements shall be addressed in the first iterations, i.e., what is expected by the grant agreement, by the task description. Any extra functionality, not expected officially, should be developed afterwards.
- Rely on the expert advice of the expert partner STFC, also in case of scope and risk management of the joint development effort.

This document briefly summarizes all the requirements we identified and discussed so far, including several “extra / delighter” ones concerning long-term future developments beyond the scope of IMPULSE. However, the next step will be to develop a roadmap and detail further the specifications by Apr 2022, in D3.8. That deliverable will also heavily consider the above risk management aspects, especially regarding (i) the development roadmap, and (ii) which requirements are part of the basic/essential category, i.e., expected by the task description.

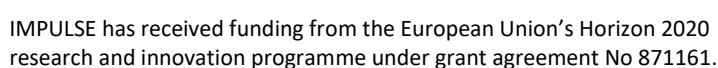
2 Conceptual approach to ELI Database of Spare Parts (EDSP)

The development of the first (test) version of the EDSP tool could be separated by functionality into several modules as summarized in Figure 2. These are sufficiently independent so that they could be specified separately and perhaps even developed separately.

Item catalogue: The development of the database should start from setting the Item catalogue. The catalogue will contain the general information about individual items (parts and equipment) categorized by type into a catalogue. Depending on the category, the items could then be searched based on relevant parameters. Having a common catalogue of parts will support the overall standardization of parts. A more detailed description of the catalogue is given later in the text.

MODULE1: This module is the core tool for identifying and classifying critical “in-system parts” that should be backed up by spare parts. In the simplest form this consists of three main data tables:

- Summary of system breakdown into subsystems and links between subsystems marking their functional interdependences - *see 2.4 below*.
- Table of “in-system parts” (with focus on key equipment) linked to individual subsystems.
- Table with “in-system parts” criticality analysis data (analysis of need) – *see 2.5 below*.



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Module 3: This includes providing the necessary documentation to support the development of a future RestAPI for the data base to facilitate a possible future ERP integration, using OpenAPI standards (this is an extra module, not really necessary for supporting the core functionalities of the spare parts database).

Module 4: Concerns developing of tools for easy reporting of data relevant for management decisions such as procurement of critical spare parts, rate of spare parts use for individual systems or sites etc. Additional functionality can be developed to generate reports on parts due for replacement or for regular inspection or calibration or as part of preventative maintenance.

2.1 Recording of equipment and parts in the database

It is clear that the assessment on which parts are the most critical to stock in order to support reliable operation of the complex laser systems must be based on knowledge of the parts (equipment) used in the system, where they are used in the systems, and what are the interdependencies between systems. In other words, we need first data that will help us to determine what will be the impact of a failure for a particular part (equipment).

It is therefore necessary to create a complete list of subsystems (at some chosen granularity) which together makes up the whole commissioned part of the facility. The list should be to some degree hierarchical (defining parent-child dependencies) but should also allow defining functional links between systems where necessary.

Having available such structure defining the sub-systems and their dependencies then allows us to link to it physical parts (equipment) used in the system as shown schematically in Figure 2. In this approach, it is not necessary to create a complete list of parts but instead, concentrate first only on what are considered key parts for a particular subsystem and gradually develop a more detailed parts (equipment) breakdown of the system.

We need to be able to record in the database, in general, any small part, equipment, instrumentation or even whole systems delivered and add suitable tags (IDs) and also add relevant parameters as those will allow us to search the records easily. Introducing a concept of an Item as a database record describing a particular part, equipment or system in terms of its category and technical parameters as shown very schematically in Figure 3.

Since we are primarily interested in recording the equipment in use in the commissioned systems and the corresponding spare parts, all items have been assigned a status based on their use. The proposed status options based on usage are the following:

- A. “Item: System part”: for parts or equipment of the system that is to be maintained, regardless of whether it is already commissioned, installed or only under construction. (Note: The status of the system will be recorded with each sub-system record.)

- B. “Item: Spare part”: for parts of equipment that are intended to be used as spare parts for the “Item: System parts”
- C. “Item: Test and measurement”: for parts of equipment that are specifically dedicated as test equipment that can be moved between systems
- D. “Item: Experimental Loan pool equipment”: for parts of equipment that are specifically intended to be for common use in experiments and borrowed for assembling different user setups.
- E. “Item: Other”: for other parts or equipment available for use – e.g., for testing etc.; unused parts from previous developments that can’t be assigned as spare parts
- F. More status options should be possible to define based on a particular need, e.g., identifying parts reserved for particular new construction.
- G.

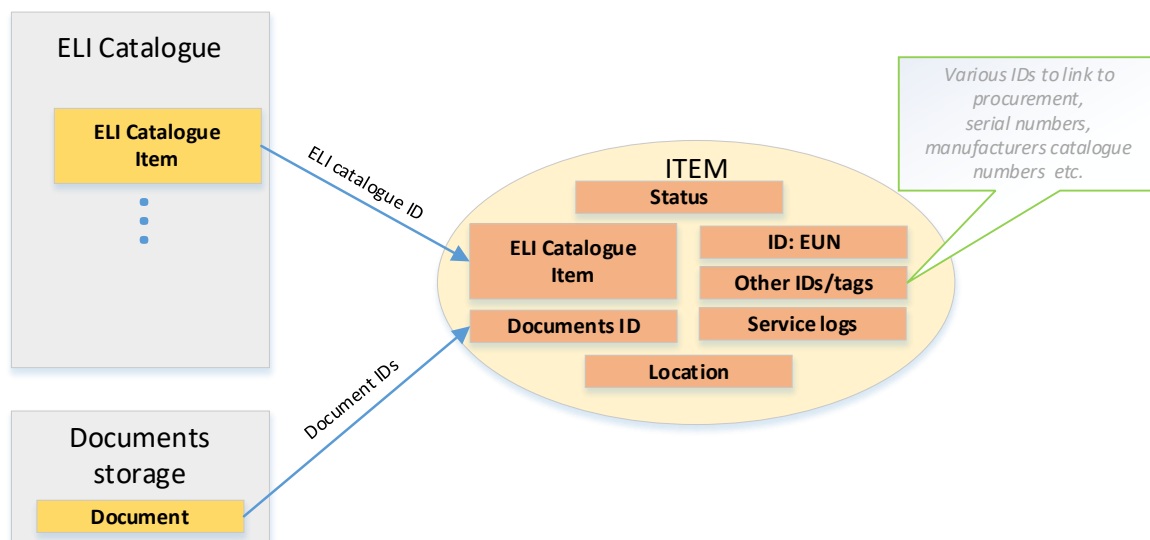


Figure 3: Concept of an Item describing any part, equipment or system

For the “Item: System part” it is then possible do the criticality analysis of needs to stock spare parts or to have service contracts and subsequently link the parts in the system to inventory of spare parts consisting of items with status “Item: Spare part” (see Figure 2 and Section 2.6) or link to existing contracts.

The records of parts or equipment with status “Item: System part” can also support regular and preventative maintenance functions in future extensions of the database.

2.2 Item catalogue

It is useful to create within the database a catalogue of items, similar to an e-shop catalogue, to keep record of each type of Item (i.e., part, equipment, or even a larger system) and to assign each unique type a catalogue number. The catalogue will contain only Items (parts) that are already owned or Items (parts) that are available on the market and can be described by a relevant set of parameters. The Items in the catalogue shall be organized into easy-to-follow tree of categories. Each category will have its own (or inherited) set of parameters appropriate for the category. The categories and the parameter definitions shall be editable within the database by a user with privileges to do so.

Each type of Item shall have not only set of technical parameters defined, but also possibility to add photo, attach files with technical sheets, 3D models, manufacturers name and manufacturers catalogue number etc.

Technically it should also be possible to include estimated item price or at least a price range (4-5 price ranges). This is still under discussion, but would be helpful for evaluation of the item criticality.

The existence of the item catalogue will significantly simplify data entry for items that have multiple instances in the system. It also aids standardization of Items used, since the database user designing of building a new system can easily find Items based on their parameters choose preferentially those that are already used elsewhere.

2.3 Use of catalogue items for individual item instances in the database

Using the catalogue number, all of these parameters and Catalogue Item related documents can then be available to any instance of the Item in the database. Individual instances of items shall then have additional information added as shown in Figure 3. This includes for instance the following, where values marked in red are the default values to be pre-filled:

(A) EUN: Unique ID for each instance of the Item consisting of Facility code + 8-digit number. Suggested facility codes:

A = ELI-ALPS

B = ELI-Beamlines

N = ELI-NP

Note 1: This ID uniquely identifies any part; can be used also for inventory purposes. Already in use at ELI-BL.

Note 2: From the first letter of the EUN one can then derive the ownership of the Item, which may be important in for example unique labelling of devices within one facility.

(B) Item status

In-system	
Spare part	
Other	
For experimental setups/loan pool	
Test and measurement	

(C) Procurement status

Planned	
Requested	
Ordered	
Delivered	
Stocked at supplier	
Destroyed/unsable	
Deleted	
No status	

(D) Status based on condition

New	
Used	
Refurbished as new	
Refurbished	

(E) Status based on serviceability

Non-serviceable	
Serviceable at ELI	
Serviceable onsite by supplier	
Serviceable at supplier	

(F) Consumable Yes/No

(G) Procurement info – delivery date, supplier, warranty expiry date, link to contract or order, invoices, delivery notes etc.

(H) Serial number

(I) Documents – links to additional documents to the documents contained in the Item catalogue, such as test and verification documents, additional photos etc.

(J) Service logs table

Notes – user notes

2.4 System structure: Record of installed serviceable equipment (Equipment breakdown)



Systems at all three facilities can be logically subdivided into subsystems with different level of detail (as needed) where these subsystems have relational links naturally to higher and to lower-level systems but also functional relations between related systems. The systems relations then describe the system structure in the facility. Each subsystem shall be described in the EDSP by additional set of attributes, such as system name or who maintains the subsystem. Additional examples of these system attributes are listed in Figure 4. Many of these attributes can then be by default propagated to related system to minimize data entry.

Physical items must always have defined subsystem to which they belong, since the subsystem carried information about location and ownership etc., but on the other hand each subsystem can only contain one physical item or no item at all.

As mentioned above the subsystems define relations to other subsystems (hierarchical parent-child, or functional allowing to determine functional dependencies between subsystems) and these relations are directional. There can be unlimited number of relations from a subsystem to other systems and it should be possible to define different types of relationship.

Attributes and parameters are inherited from parent systems but can be user edited.

Defining who is responsible for the subsystem can be perhaps also used for assigning editing rights in the database.

In this representation the sum of “children” subsystems does not have to equal the parent subsystem, i.e., not all parts of the installed systems have to be listed in the database.

Physical items can be easily moved within the system structure by assigning them to a different subsystem or by changing the relations between subsystems. This will enable simple recording of exchange of in-system Items for spare Items.

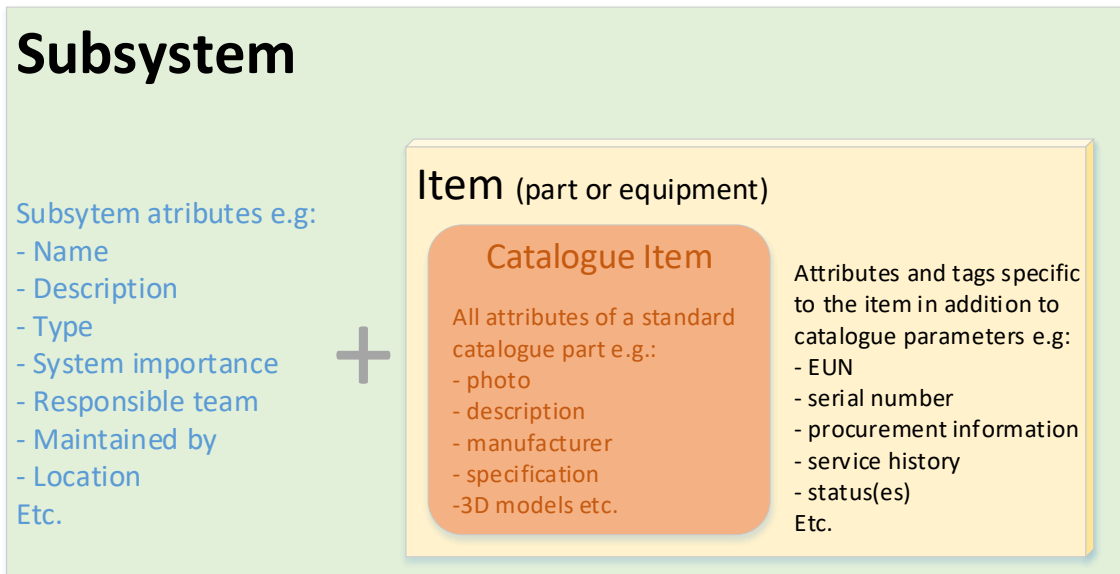


Figure 4: Concept of representing the physical item by its catalogue parameters, by additional individual item tags and by its location within the system structure.

In summary any physical part is then described by three layers of information that together gives efficiently the full description of the parameters, identification, history, use, location etc.:

- 1) Fixed information for the Item catalogue common to multiple items
- 2) Specific information on individual physical Item including its status and history
- 3) Information on the subsystem where the Item is used including
 - (A) **Information on relation and functional dependences on other systems**
 - (B) **Information on ownership and responsibility for the subsystem**
 - (C) **Location** – information on where the part is located, e.g., warehouse shelf number, or where the part is located within the system.
 - (D) **Tags for CS** – e.g., C059 to denote camera number independent of the physical item. Multiple tags associated with subsystem possible for different use.

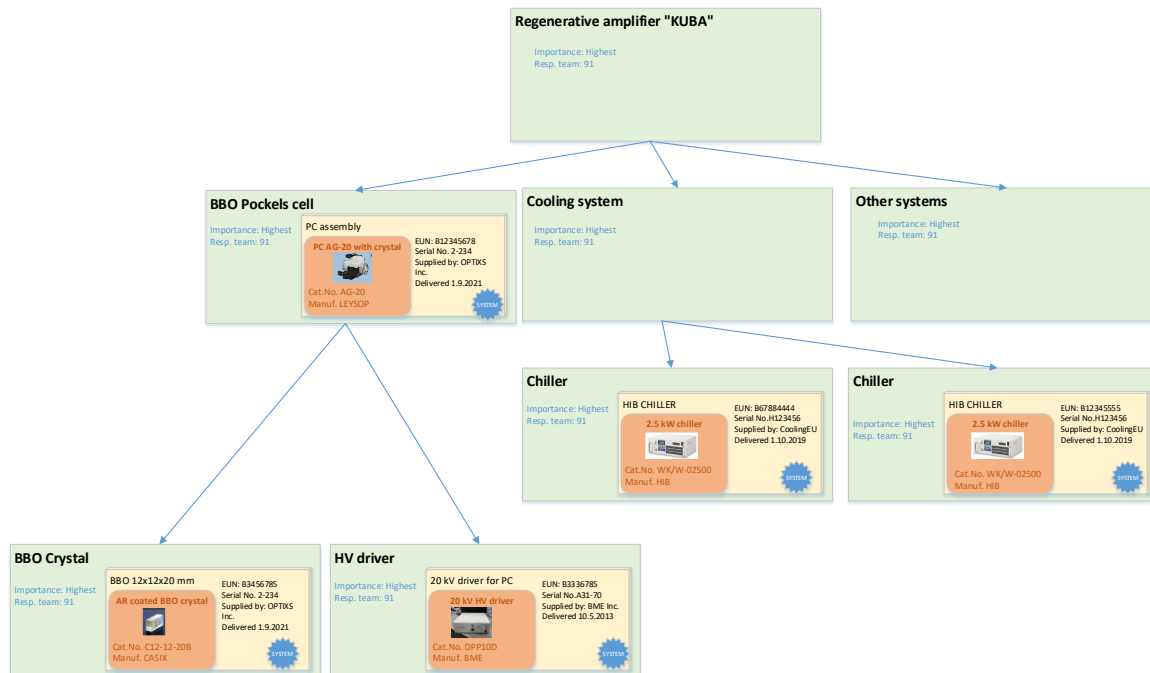


Figure 5 An example of simple system structure demonstrating concept of subsystems each containing one item or no items.

2.5 Spare parts criticality analysis

The need to evaluate and classify the spare parts based on their criticality in the laser and experimental system operation is the key feature of any spare parts inventory system. There are many factors that enter into consideration which spare parts are the most critical and valuable to have in stock ready to be installed in the system, be it based on impact on the operation, logistical factors (cost, lead time, shelf lifetime, number of available suppliers) or maintenance-based factors (replacement time, replacement difficulty, failure rate etc.). These factors need to be systematically evaluated to determine how critical the particular part is for the operation and for successful completion of user experiments. The evaluation must ideally provide a single number that will determine criticality of the part and in a simplified form classification of the part into one of four criticality classes (categories). Additionally, and analysis of the instances of the part in the system combined with the determined criticality class should guide a decision of how many parts should be stocked. Suggested criticality classes are listed in Table 1.

Development of the criticality analysis formula (and which factors to include and with what weight) is the first part of the EDSP that should be developed. The identification of critical spares based on criticality class is essential for ensuring reliable and safe operation of any complex system. Furthermore, the criticality class can serve as the basis preferential procurement and determination service requirements as well as for training of staff in particular maintenance skills.

The criticality analysis functionality should be ideally developed together with the system structure (interdependence) and equipment breakdown (ref. to 2.4) in order to determine the factors linked to the impact on the system.

Here follow some thoughts on factors that enter into consideration in the analysis [1], [2]. In case of part failure, the logistical factors that determine the criticality of the spare parts and have therefore been considered in the evaluation are the following. From the maintenance point of view, it is the importance of the system where the part is used.

Impact criteria:

- System function – from most important to least important – *inherited from system structure, see section 2.4*
- Equipment criticality within system]
- Type of impact – no impact on experiments, quality impact on experiments, severe reduction of experimental work, complete stop of experiments.

Part lead time criteria– consisting of two components

A) Mean Logistics Delay Time (MLDT) – this includes administrative delays in ordering the equipment (can be mitigated to some degree – e.g., by setting up framework contract with suppliers), but we could count here also delays in bringing in qualified service personnel for the repair.

B) Delivery time from order (DTFO) – time needed to deliver the part from date of placing the order.

Number of potential suppliers' criterion – how many suppliers are available on the market, the lower the number the higher the risk and consequently the higher the criticality.

Failure rate per year (or 1/lifetime) of the part/equipment.

Remaining useful time (RUL) of the part or equipment in the system that the spare part is supporting. Parts with that have low RUL should have higher priority for stocking the spare part.

Mean Time to Repair (MTTR) – time needed to identify the exact problem, repair the damage equipment and restore the operation back to normal working condition (includes diagnostic time, alignment time, testing time, software installation and setup etc.). This is important to enter into database, but it does not enter into criticality evaluation as it only determines minimum downtime in case of failure. It would be important, however, if we considered preventative maintenance in future.

Table 1: Spare part criticality class (adapted from [2])

Criticality Class	Description
1: High	Unavailability of the part causes unacceptable condition, either operation or with respect to safety, e.g., mandates complete stop of the user experiment Lead time or risks in ordering process are very high
2: Medium	Unavailability of the part causes unacceptable condition, but the consequences can be partly mitigated, e.g., user experiments can be done with reduced laser parameters Parts need to be supplied within a short time period A calculated risk in the lead time
3: Low	Unavailability causes an acceptable condition, e.g., motion stage has to be moved manually instead though motion control. Lead time risk is acceptable
4: No	Unavailability has no effect on operations, personal safety or machine safety, e.g., broken information panel where data can be displayed elsewhere Risk in process of ordering is normal

The determination of the criticality class for a particular spare part could be based on assigning points to each of the relevant criteria and then adding up the point contributions with weighting factors for each criterion to get criticality indicator value CP :

$$CP = w_1 C_1 + w_2 C_2 + \dots + w_n C_n$$

Where C_1 to C_n are points assigned to individual 1...n criteria and where weighting factors w_1 to w_n can be adjusted based on importance of individual criteria. Appropriate intervals of CP values can then be assigned to the four values of the criticality class.

This is the simplest form of evaluation of spare parts criticality but more sophisticated evaluation can be developed later based on the already existing data.

2.6 Inventory of spare parts

The inventory of spare parts (Module 2) shall be a subset of inventory of uniquely identified parts (items) that have a status of “spare part”. Ideally, these should have identified link(s) to one or more in-system parts for which they are reserved as spares. The types of links and the process and rules for setting them up has to be still discussed and defined. The support of the in-system parts with spares should be evaluated by the criticality analysis described above with, for example, insufficiently supported in-system parts carrying high risk highlighted in color.

2.7 Identification of common spare parts and joint procurement



Especially in cases where criticality analysis concludes particular in-system part is insufficiently supported by spare parts there must be defined a way of either requesting either purchase of spare part or requesting some existing part to be linked as spare to the in-system part. The request for purchase should be ideally linked to ERP system and after approval should be purchased and item should then appear in the EDSP as a spare item but with a procurement status (e.g., “requested” and date e.g., 10th Oct 2021) and its procurement status should automatically change based on the procurement progress.

3 Technical approach to ELI Database of Spare Parts (EDSP)

3.1 Interfaces to existing data

The existing data sources at each ELI site should ideally be migrated to the new EDSP database preferably in one step or linked through appropriate API to the database gateway. The programming of the data migration will have to be done locally.

It will be a synchronization service, that will follow the base principles of ETL (extract, transform, load). ETL is the general procedure of copying data from one or more sources into a destination system that represents the data differently from the source(s) or in a different context than the source(s). On the side of EDSP the API Gateway will provide all necessary endpoints to satisfy this approach. We will use strong identifiers (GUID) and timestamp properties on both sides which help us to easily keep both sides updated.

3.2 Security

The system will be accessible only for authenticated users with the right set of authorization.

For now, authentication will be implemented via local database user and password. Later we will use the LDAP of ELI-ERIC to authenticate users.

The authorization will be role-based, with an authorization system that uses dynamic entities (e.g., DB table, special GUI button, ...), where we can define access for four base actions, create, read, update and delete, also known as CRUD operations.

Example of roles and access rights:

possible roles and entities	Catalogue (subentities: codebook, catalogue)	Physical Items	Subsystems (sub-entities:
-----------------------------	---	----------------	------------------------------

	category...		relationships
system admin	CRUD	CRUD	CRUD
catalogue editor	CRUD	R	R
Item editor	R	CRUD	R
System structure editor	R	R	CRUD
Viewer	R	R	R

3.3 Interfaces to ERP system

It is expected that all three ELI sites will in the future use a common ERP consisting of multiple modules (finance, procurement, human resources, warehouse management etc.) which will mutually exchange data through some standard API interface. The EDSP could eventually become one of these modules closely integrated with the ERP.

Since the ERP system for ELI is being developed in parallel, it is at this point impossible to define all of the requirements for the integration. Instead, we will initially develop that EDSP as a modular stand-alone isolated system, structured in the way that it can be later connected to “standard” ERP modules such as procurement, warehouse management etc.

3.4 Proposed technical architecture

We will use an architecture based on a centralized API Gateway that will handle all accesses to the database as shown blow in Figure 6.

All external interfacing to the database data will be done without exceptions through this centralized gateway behind which will be individual microservices, each covering a specific area (e.g., catalog service). The centralized gateway will be based on OpenAPI REST API standard and this will make it extremely flexible and easy to interface to other data sources and in future also integrate ERP system modules or migrate any local data compatible with OpenAPI REST API. The same gateway will be also used develop front-ends (ERIC webapp GUIs for Spare parts, System structure, maintenance functions...).

This architecture will help us to keep the system and the development sustainable and scalable.

We will use a relational database for first prototypes. Because of the complexity of the subsystem hierarchy and subsystems relations, we could use some hybrid solution with graph database for example.

For data analysis and reporting, we will use a data warehouse solution or some distributed search and analytic engine.

ELI-ERIC database of systems items and operations system architecture

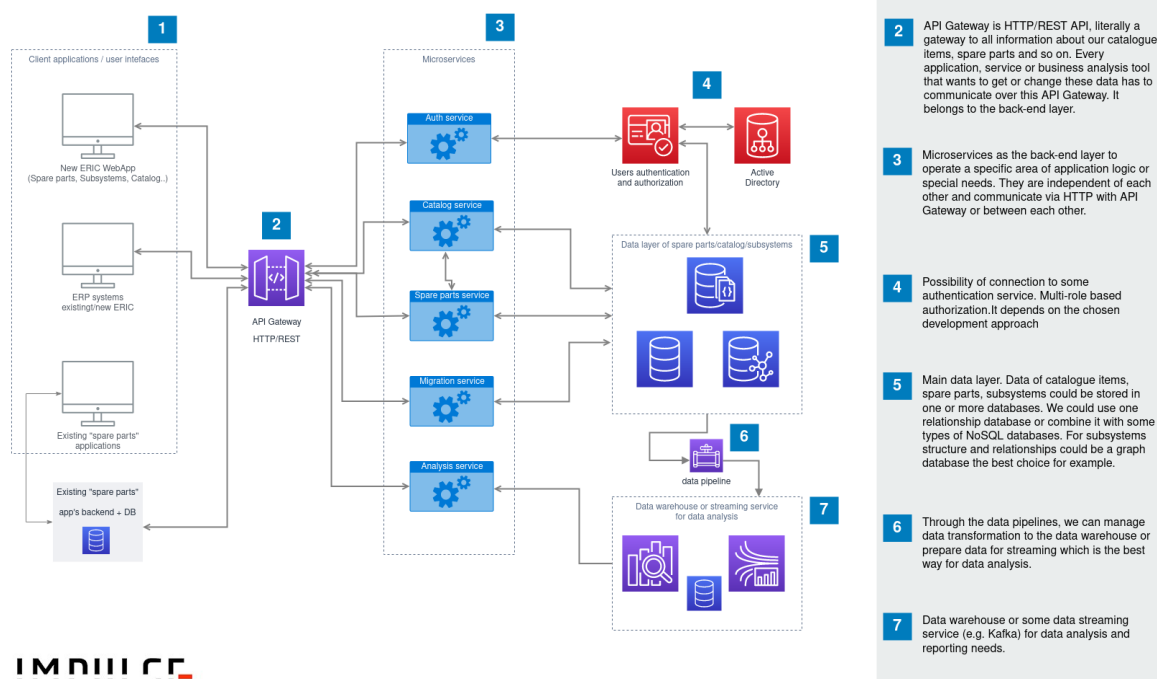


Figure 6: Proposed software architecture for implementation of the EDSP.

3.5 Visual Prototype, Mock-up

The development of the test database will start with writing parts of the code that is needed for the implementation of the described architecture in Figure 6. At the same time decided we will also start preparing the database user interface and develop ideas on how to make this interface easy to use and intuitive. To help us to do this we have started using the prototyping tool proto.io to create non-functional GUI mockups. In this proto.io tool, we can easily collaborate on front-end design. We also expect to do more sophisticated mockups and functional prototypes using javascript framework based on React. An example of the od mockup screen with a selection of different types of optics is shown in Figure 7.

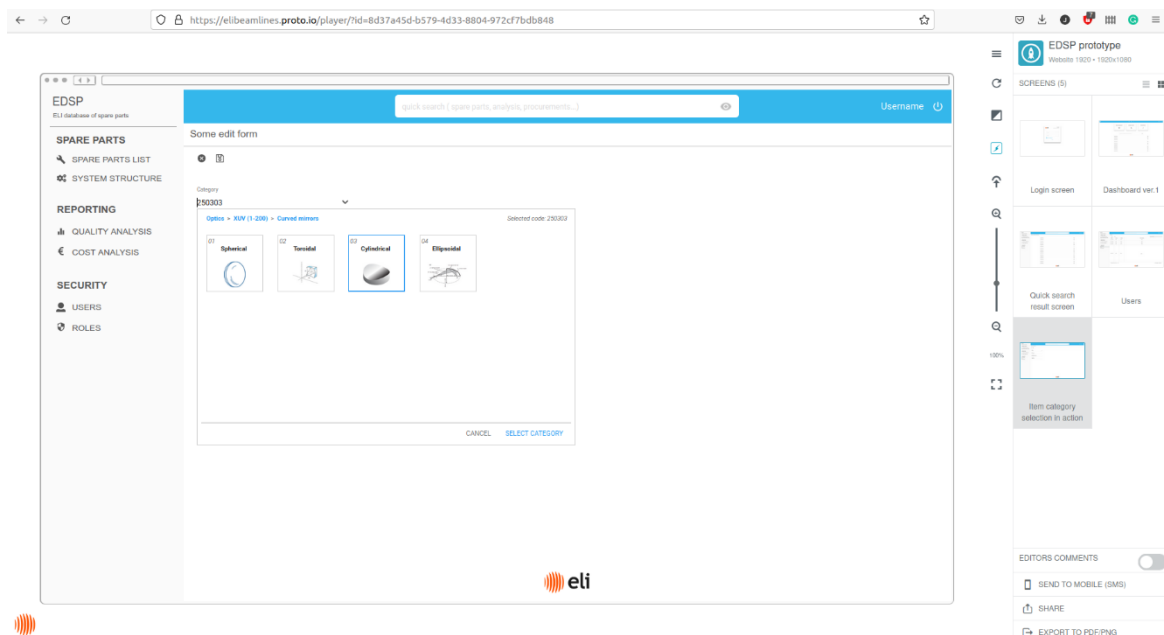


Figure 7: An example of the prototyping tool screen showing selection of item categories

4 Requirement specifications

In addition to the summary of the concepts agreed for the developments of the EDSP we summarize here the formal numbered requirements. These can be used to verify the final software product.

4.1 Functional requirements

TF 3.3.1 System analysis [major scientific equipment analysis]

1. The system shall support the identification of spare parts, through alternative ways:
 - 1.1 Through the flexible definition of criticality functions
 - 1.2 Through the list of critical items (if available)
 - 1.2.1. The system should allow recording the list of critical items via the GUI
 - 1.2.2. The system should allow the recording of items by a group of people (laser & beamline teams, multiple users)
 - 1.2.3. The system should allow to import the list of critical items from an Excel file (according to a pre-defined format/template)
 - 1.3 Through the analysis of the previous year(s) spare part usage
 - 1.4 Or any combination of the above

2. The system should be flexible enough to support the identification of parts in general (i.e., not only spare parts but e.g., user part, pool part, development part, built in part) through alternative ways:
 - 2.1. Through the list of relevant items (if available)
 - 2.1.1. The system should allow recording the list of relevant items via the GUI
 - 2.1.2. The system should allow the recording of items by a group of people (laser & beamline teams, multiple users)
 - 2.1.3. The system should allow importing the list of relevant items from an Excel file (according to a pre-defined format/template)
 - 2.2. Through the analysis of the previous year(s) part usage
 - 2.3. Or any combination of the above
3. The system should allow the management of parts in a general and integrated manner (a part may change from spare to user part upon request)
4. The system should allow the management of loan parts (& related process), i.e., requesting and using parts from another lab and procuring another replacement one for this (not core functionality, but should be extended towards this)

TF 3.3.2 Optimization of spare parts procurement and standardization

1. Standardization of equipment (component) / support ELI-wide standardization (GA)
 - 1.1. The system shall support different level of identification systems:
 - 1.1.1. Part serial number: unique identifiers for all the physical parts, as provided by the manufacturer [manufacturer-specific]
 - 1.1.2. ELI inventory number: unique identifiers for all the physical parts (i.e., for each physical part) [harmonized within ELI]
 - 1.1.3. Manufacturer part number: unique identifiers for all the parts with the same specification (i.e., for each logical part), from a specific manufacturer
 - 1.1.4. ELI part number: unique identifiers for all the parts with the same specification (i.e., for each logical part), independently from the manufacturers
 - 1.1.4.1. The system shall include a built-in algorithm for crosschecking the database whether a new logical part is already there [and instead of registering a new part number redundantly, proposing the already available part number]
2. Allow a more strategic approach to procurement / optimize procurement / joint procurement planning (GA)
 - 2.1. ELI-wide procurement of similar spares (GA)

- 2.1.1. Upon the related Facility reports collected [depends on TF3.3.3 + TF 3.3.4] the system should be able to generate a report on top of these, about similar spares (in the appropriate format / towards the prescribed interface - depends on WP2 / ERP]
- 2.2. Joint management of stock (GA)
 - 2.2.1. The system shall allow searching for potentially available spare parts in the other Facilities [depends on WP2, mgmt. preconditions for this / what is potentially available to share, etc.]
- 2.3. The system should be flexible enough to support the technical aspects of the procurement of any parts in general (i.e., not only spare parts but e.g., user part, pool part, development part)
- 2.4. The system should allow the user to split the procurement needs according to the suppliers' capability to deliver (e.g., a supplier is able to deliver cameras, filters and mounts but cannot deliver vacuum), also to be in compliance with the procurement rules (e.g., to have 3 bidders at least)
- 2.5. The system may even assist the user to split the procurement needs according to the suppliers' capability to deliver (i.e., based on former procurements/supplier database)

TF 3.3.3 Local adaptation and implementation at ELI Facilities

- 1. Systematic data collection on spares & technology components (GA)
 - 1.1. The system should allow to record the results of systematic data collection about laser & secondary source systems, end stations, and support systems [e.g., Optical workshop Task 4.1], according to the jointly predefined format and aspects [first step could be: prepare a template table for this, collect the current status for all systems]
 - 1.2. The system should allow to record and/or integrate the results of systematic data collection about all related local interface and templates
 - 1.2.1. ELI-ALPS part number structure [full export to Excel]
 - 1.2.2. ELI-ALPS procurement initiation form (Excel template)
 - 1.2.3. ELI-ALPS take-over/delivery form (Excel?)
- 2. The system should fulfill the high-level requirement of: effectively collect, manage and integrate all available information on spare parts (GA)
- 3. The system should be flexible enough for local specificities, especially for the followings:
 - 3.1. The system should allow the management of any parts in general (i.e., not only spare parts but e.g., user part, pool part, development part)

- 3.2. The system should support multilingual attributes supporting multilingual procurement/stock management/reporting (especially to have everything in one system, related to local procurement rules and local reporting needs)
- 3.3. The system should support importing online shopping lists (e.g., through a browser plugin, at least the possibility of such extensions should be considered and addressed)

TF 3.3.4 Reporting and extraction features for analysis

- 1. The system will be a core source of information for regular review meetings leading to the definition and evaluation of operational modes from an operational risk perspective (GA)
 - 1.1. The system should provide a statistic of the rate of spare parts consumption and replacement (GA)
 - 1.1.1. The system should follow the lifecycle of spare parts (consumption, replacements)
 - 1.1.2. The system should allow the possibility of following the lifecycle of parts in general (user part, pool part, built-in part, etc.)
 - 1.2. The system should provide the rate of spare parts consumption and replacement will be analysed relative to the selected laser performances (GA)
 - 1.3. The system should be able to analyze (or support the analysis) of the impact on the laser systems uptime on the rate of spare parts consumption and replacement (GA)
 - 1.4. The system shall be able to export reports for next year OPEX planning [final format & template to be aligned and discuss with WP2]
 - 1.5. The system shall be able to export reports for 5-year OPEX planning [final format & template to be aligned and discussed with WP2]

TF 3.3.5 Joint development of spare parts database

- 1. The system should be specified jointly, implemented at each Facility separately (GA), however, the final agreement was to implement the common central parts jointly, and only the local data migration parts separately. See also a dedicated paragraph about this in the Introduction.
- 2. The system should allow user defined functions
- 3. The system should have multilingual support in general
- 4. The system shall provide a web-based user interface
- 5. The system shall provide a command-line interface as a redundancy

6. ...

4.2 Non-functional requirements

TF 3.3.5 Joint development of spare parts database

1. The core designs (and developments) shall be harmonized among the ELI Facilities.
(GA)

1.1. The system shall be developed in well-known and standard programming languages, environments, frameworks and tools. The architectural diagram is shown in Figure...

1.1.1 Backend (logic) development: preferred database in Postgres DB, but investigate Neo4J as an alternative. Running on-prem or in the cloud.

1.1.2 Business logic layer development: using the microservices architecture, including API Gateways as a single-entry point for all logic and microservices to implement the functionality. The API Gateway is described using the Open API standard. Microservices are written in NodeJS and Python languages.

1.1.3. Frontend (user interface) development preferred in React running on-prem or in the cloud.

1.1.4. Deployment architecture preferred on GitHub using GitHub Actions

1.1.5. Preferred integrated development environment is VS Code

1.1.6. Preferred version control tool is git hosted on GitHub

1.1.7. Preferred continuous integration approach and framework is also provided by GitHub

1.1.8. Preferred testing approach is Postman for unit testing and documented test cases for the GUI.

1.2. The system shall be developed according to the Agile principles

1.2.1 Make use of Jira and its agile concepts

2. The system shall be flexible enough for local adaptations and extensions

2.1. The system shall have a plug-in-based architecture (especially because the efficient adaptation at Facilities in TF3.3.3)

2.1.1. The system should allow user defined functions via pull requests

2.2. The system shall allow the possibility of specific and flexible APIs

3. The system shall satisfy general technical requirements

3.1. The system shall support the login with ELI credentials

3.2. The system response times should be as fast as possible. In cases where the computer cannot provide fairly immediate response, continuous feedback should be provided to the user in form of a percent-done indicator for example.

- 3.3. The system should have a microservices based architecture
- 3.4. The system shall have a redundant architecture
- 3.5. The system shall be backed-up on a regular basis

5 References

- [1] K. Antosz and R. M. C. Ratnayake, "Spare parts' criticality assessment and prioritization for enhancing manufacturing systems' availability and reliability," J. Manuf. Syst., vol. 50, pp. 212–225, Jan. 2019, doi: 10.1016/j.jmsy.2019.01.003.
- [2] A. Molenaers, H. Baets, L. Pintelon, and G. Waeyenbergh, "Criticality classification of spare parts: A case study," in International Journal of Production Economics, Dec. 2012, vol. 140, no. 2, pp. 570–578, doi: 10.1016/j.ijpe.2011.08.013.